Year 13 Further Maths - Series and induction

Similar questions - Proof by induction

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 Let $y = xe^{2x}$. Prove by induction that, for all integers $n \geqslant 1$, $\dfrac{d^n y}{dx^n} = 2^{n-1}(2x+n)e^{2x}$. **[6]**

2

(a) A sequence is defined by $u_1 = 3$ and $u_{n+1} = 2u_n - 1$. Prove by induction that $u_n = 2^n + 1$ for all positive integers n. **[5]**

(b) Prove by induction that $3^{2n} - 1$ is divisible by 8 for all positive integers n. **[5]**

Question 1

$n=1$: $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \mathrm{e}^{2x} + 2x\mathrm{e}^{2x} = (2x+1)\mathrm{e}^{2x}$, which matches $2^0(2x+1)\mathrm{e}^{2x}$. Assume true for $n=k$: $y^{(k)} = 2^{k-1}(2x+k)\mathrm{e}^{2x}$. Differentiating: $y^{(k+1)} = 2^{k-1}\left[2\mathrm{e}^{2x} + 2(2x+k)\mathrm{e}^{2x}\right] = 2^{k-1}\mathrm{e}^{2x}(4x+2k+2) = 2^k(2x+k+1)\mathrm{e}^{2x}$, the formula for $n = k+1$. True for $n=1$ and true for $k \Rightarrow k+1$, so true for all integers $n \geqslant 1$

Question 2

(a) $n=1$: $2^1+1 = 3 = u_1$. Assume $u_k = 2^k+1$. Then $u_{k+1} = 2(2^k+1)-1 = 2^{k+1}+1$. True for $n=1$ and $k \Rightarrow k+1$, so true for all positive integers n

(b) $n=1$: $9-1=8$, divisible by 8. Assume $3^{2k}-1 = 8m$. Then $3^{2(k+1)}-1 = 9\times 3^{2k}-1 = 9(8m+1)-1 = 72m+8 = 8(9m+1)$, divisible by 8. True for $n=1$ and $k \Rightarrow k+1$, so true for all positive integers n